Fear, Safety, & Reward Responder Subtypes and Skin Conductance Response Predictions

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INTRODUCTION

- Previous studies demonstrate that PTSD, among other disorders, is associated with difficulty reducing fearful responses in the presence of safe stimuli (Fitzgerald & Webb et. al., 2025).
- Self-report measures on PTSD, Alcohol Use, Depression, and Anxiety are used to inform diagnoses and are often used in combination to assess comorbidities (Levitt et. al., 2021).
- Hierarchical clustering techniques allow for similarity between all patient responses and offers novel insight around diagnostic stratification before treatment assignment.
- Developing a predictive model for each diagnostic-subtype would inform an individual's likely responses across a Fear, Safety, and Reward exposure task.

METHODS

- N = 128 participants were recruited through the Psychology department subject pool.
- Questionnaires on Trauma (PCL-5, CAPS-5, CTQ), Anxiety (IUS), Alcohol Use (AUDIT), Depression (MINI MDD Subscales), and SSRI Use were administered.
- Participants had electrodes placed on their fingertips to collect skin conductance responses (SCR) and completed the FSARS task, where 5 colored geometric shapes/outlines are presented 8 times in random order across 40 trials.
- Unconditioned stimuli (US) such as aversive noise (Fear), $0.25 (Reward), or no result (Neutral) were linked with geometric shapes (conditioned stimuli [CS]). (Figure 1)
- Gower's distance was computed for participant's questionnaire responses and Agglomerative (ANGES) clustering was applied using Ward's linkage.
- Banner plot inspection revealed a 6-cluster solution with common response types ranging from low-risk to high-risk clusters. (Figure 2)

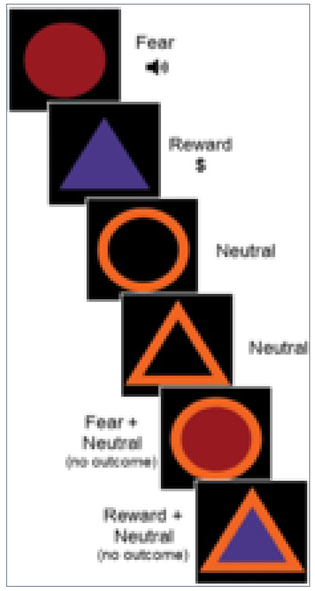


Figure 1. *Presented stimulus examples*

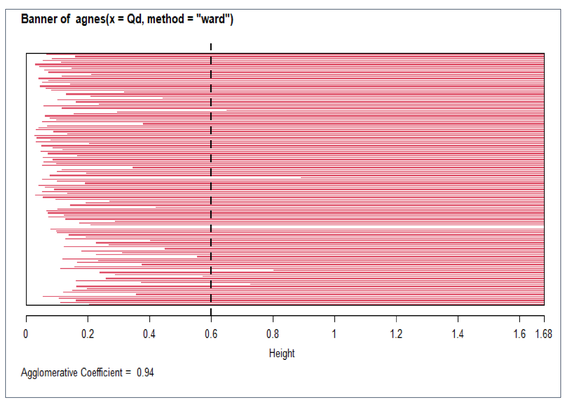


Figure 2. *AGNES Revealed Cluster Banner Plot.* Dotted line at most clusters with acceptable stability

- A Naïve Bayes classifier was trained on 70% of participants and validated on 30% with psychometric features as predictors as a method to classify new individuals without requiring physiological data.
- Participant cluster membership was merged with trial level SCR data to reveal each cluster's habituation trajectory across each stimuli type.
- LOESS-smoothed trendlines were plotted for SCR response as a function of conditioning trial number within each cluster-outcome combination
- To model the temporal dynamics of psychophysiological responding and generate forward projections, Holt double exponential smoothing models were estimated on 75% of each trial-averaged, log-transformed SCR time series, with the remaining 25% withheld for out-of-sample validation. Fear+safety responses were the primary focus of this analysis.

RESULTS

- Cluster 5 represented the highest-burden archetype, exhibiting the greatest PTSD severity (PCL M = 47.3; CAPS-5 M = 5.29), intolerance of uncertainty (IUS M = 49.1), and rates of current, past, and recurrent MDD (43%, 86%, and 43%, respectively), alongside the highest SSRI use (57%).
- Cluster 2 emerged as the lowest-symptom archetype (PCL M = 14.3; CAPS-5 M = 1.38; IUS M = 22.6).
- Cluster 6 was distinguished by the highest alcohol use (AUDIT M= 8.64) and a 100% rate of past MDD with minimal current symptoms.
- Naïve Bayes revealed a 91.67% accuracy in classifying observations to clusters based on self-report measures.
- Non-linear trends reveal that clusters 1, 2 & 4 have steadily decreasing or stable SCR to reward and/or safety, but stable near-zero SCR to fear.

Measure	C1 Mod. PTSD / Anxiety	C2 Low symptom	C3 Trauma / MDD hist	C4 Mild / All women	C5 Highest symptom	C6 High alcohol
PCL	20.8	14.3	22.7	15.4	47.3	22.0
CAPS-5	3.53	1.38	4.14	1.32	5.29	3.09
CTQ	31.9	33.9	43.3	35.9	43.7	41.0
CTQ-EA	7.34	8.12	11.1	8.84	11.0	10.2
IUS	38.0	22.6	28.1	27.6	49.1	26.3
IUS-Pro	24.2	14.9	18.5	18.4	28.9	15.5
IUS-Inh	13.8	7.71	9.64	9.16	20.8	10.6
AUDIT	7.07	5.59	4.50	3.56	4.71	8.64
MDD Current	0%	1%	19%	1%	43%	1%
MDD Past	1%	5%	88%	9%	86%	100%
MDD Recurrent	1%	3%	100%	4%	43%	2%
SSRI Use	24%	24%	19%	24%	57%	9%
Age (M)	19.3	18.7	19.4	19.0	19.1	19.1
Gender	Mixed	Mixed	Mixed	All Female	Mixed	Mixed

Table 1. *Cluster Means on Self-Report Measures Across Six Psychologically Derived Archetypes.* Highlighted badges denote the highest and lowest values for each measure across clusters.

- Clusters 3, 5, & 6 have decreasing or stable SCR by trial in response to fearful stimuli with humps around trials 4 or 5. Responses to reward were largely neutral to rewarding stimuli.
- Time series Holt's smoothing across 40 trials predicts a trauma response increasing for Clusters 1 & 5, neutral for Clusters 4 & 6, and decreasing for Clusters 2 & 3 in forecasted trials. (Table 2)

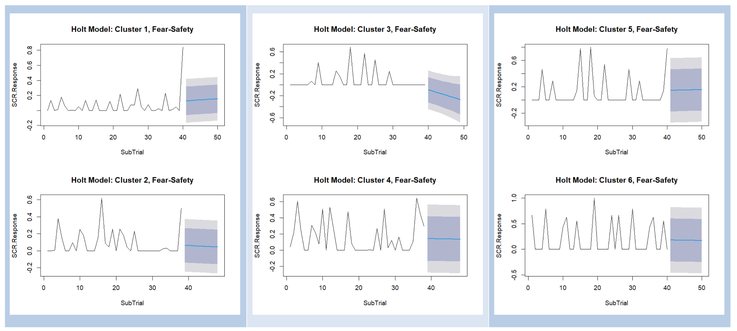


Table 2. *Holt Exponentially Smoothed Time Series Forecasts by Cluster Means*

- Training set accuracy was good (MSE = 0.08, MASE = 0.88). Test performance was comparable (MASE = 0.92), with small error (ME = 0.22) indicating modest underestimation of SCR magnitude in held-out trials. Percentage-error metrics were undefined due to zero-valued observations and are not reported.

CONCLUSION

- Hierarchical clustering on self-report measures produces psychologically coherent archetypes Autonomic response trajectories across a fear-safety-reward paradigm are both distinct and forecastable.
- Archetype-informed treatment matching has potential to meaningfully improve intervention precision for PTSD and comorbid conditions.
- Future research to improve upon individualized treatment plans will focus on the incorporation of Bayesian Hierarchical Time Series models. The partial pooling involved would allow estimates to be made of individual's trajectory and cluster-level trajectory simultaneously.

REFERENCES

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